

Functional Requirement Document (FRD)

Interactive Product Customization Engine for Golf Apparel

Project Title:	Ka.Sha Golfwear T-Shirt Configurator E-Commerce Integration	Document Version:	v1.0
Target Audience:	Full-Stack Web Development Vendor / UI Developers	Date Created:	May 23, 2026

1. Document Purpose & Global System Architecture

This document details the functional specifications and user interface layouts required to build an interactive, multi-step golf t-shirt customization wizard. The implementation must map strictly to the core logic provided in the reference hand-drawn design layouts (Ref: *WhatsApp Image 2026-05-23 at 12.16.18.jpeg*).

1.1 Global State Rules & Persistent Controls

- Persistent Navigation Footer:** Every step of the customization workflow must display a fixed bottom navigation bar featuring three critical actions: [← Back], [Continue to Next Step →], and [🛒 Add to Cart].
- State Retention Cross-Navigation:** If a customer changes their layout options or returns to a previous stage (e.g., from Step 3 back to Step 2), all current canvas modifications, design selections, texts, and uploaded files must be maintained in state without a reset occurring.
- On-Screen Instructions:** Simple, prominent instructions must be permanently positioned above the interactive panel on each tab view to ensure zero-friction usability.

1.2 Birds-Eye Progress Tracker Layout

A persistent progress pipeline must sit at the top of the system viewport, clearly illustrating the workflow stages:



2. Tab-by-Tab Technical Specifications

Tab 1: Choose Product Style

On-Screen Customer Instruction:

"Start your design! Select a foundational style for your golf t-shirt from the options below to begin customizing."

Functional Requirements:

- Display a horizontal, mobile-responsive grid configuration listing foundational t-shirt patterns.
- Available configurations to expose: **Solid**, **Print**, **Pattern 1** (Color-blocked elements), and **Pattern 2**.
- Clicking on any style thumbnail must instantly reload the foundational asset mesh inside the live 2D/3D canvas viewer.

LIVE T-SHIRT PREVIEW CANVAS
(Updates dynamic style silhouette template)

CHOOSE YOUR STYLE

[SOLID]

Plain base body

[PRINT]

All-over print layer

[PATTERN 1]

Contrast body panels

[PATTERN 2]

Accent shoulder cuts

Tab 2: Select Colour / Print

On-Screen Customer Instruction:

"Color your canvas. Click on a garment section to choose a vibrant solid color or a premium print pattern."

Functional Requirements & Logic Mapping:

- **Conditional Logic Rule A (Solid / Print styles):** The dashboard displays an "Apply to All" option. Selection targets the full surface area globally.
- **Conditional Logic Rule B (Pattern styles):** Customization selections are bundled together as unified accent layers to prevent UI layout over-complication. Individual components are mapped strictly into two structural groups:
 1. **Base Body:** Controls the main foundational area of the t-shirt.
 2. **Pattern Elements:** Selects and applies one uniform pattern asset or color across all accent trims, side panels, and collars simultaneously.

LIVE CANVAS ENGINE (Reflects color/print mappings in real-time)	CUSTOMIZE DESIGN ELEMENTS If Solid/Print Chosen: ☒ Entire T-Shirt [🎨 Choose Color/Print] If Pattern 1/2 Chosen: ☐ Base Body [🎨 Choose] ☐ Pattern Panels [🎨 Choose]
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2.1 Selector Modal Pop-up Specifications

Triggering any choice button opens a clean Modal interface overlay. Two variant modal views must be supported natively:

Color Palette Pop-up Modal View

Select Common Colors:

☐ White ☐ Black ☐ Navy ☐ Green ☐ Red

Advanced Color Wheel Selector:

[Interactive Spectrum Canvas]

Color Picker Tool:

[ Sample Eyedropper API]

Cancel

APPLY

Print Swatch Gallery Pop-up Modal View

Select From Premium Print Patterns:

[Print Grid Item 1]

[Print Grid Item 2]

[Print Grid Item 3]

[Print Grid Item 4]

[Print Grid Item 5]

[Print Grid Item 6]

Cancel

APPLY

Tab 3: Add Logo & Text Customization

On-Screen Customer Instruction:

"Make it uniquely yours. Upload your team logo or add custom text. Drag and resize your assets freely within the highlighted chest area."

3.1 Logo File Management

- **Pre-fixed Boundary Logic:** The user-uploaded asset must scale and render inside a designated, secure rectangular canvas overlay matrix positioned on the chest section. Drag-and-drop constraints must prevent assets from bleeding outside this pre-fixed bounding safety zone.
- **Background Removal Tooling:** Include a discrete, one-click [✨ Remove Image Background] UI processing toggle leveraging programmatic asset mask extraction tools.

3.2 Custom Text Parameters

- Provide custom entry lines supporting structural styling transformations:
 - **Font Engine:** Fixed dropdown list option selector populate with clean, modern corporate and sporting typography presets.
 - **Sizing Adjustment:** Granular linear slider component to handle proportional text dimensions.
 - **Color Swatch Selection:** Direct routing into the native core palette picker interface.
 - **Alignment Rules:** Toggle button structures providing Left, Center, and Right justifications.

LIVE CANVAS EDITOR

Fixed Logo Box

Drag & resize bounds restriction

ADD BRAND LOGO

↑ Upload Asset Image

✨ Remove Image Background

ADD TEXT ACCENTS

Font: Sport Sans Bold Size:

Color: [🎨 Choose] Align:

L

C

R

Tab 4: Sizing & Order Matrix

On-Screen Customer Instruction:

"Select your sizing preferences and enter the quantity needed for each size to complete your order."

Functional Requirements & Checkout Rules:

- **Selection Toggles:** Allow swift configuration switches between **Standard Sizing Grid** and **Custom Dimensions Form Templates**.
- **Quantity Matrix Configuration:** Modeled explicitly after bulk distribution e-commerce architectures, customers are presented with an inputs grid enabling parallel quantity tallies across size blocks simultaneously.
- **Bulk/Custom Notice Callout:** A clear, unmissable text callout module must process routing options towards manual sales desks for specialized sizing or bulk volumes.

QUANTITY ORDER BREAKDOWN

Mode Selection: ☒ **Standard Wholesale Sizes** ☐ Custom Specification Form

SIZE CATEGORY	S	M	L	XL	XXL	RUNNING TOTAL
INPUT QUANTITY	10	15	5	0	0	30 Pcs

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